

SmartTrack

School Bus Tracking System

35-Day Development Timeline

Backend | Web Frontend | Mobile Apps - Complete Delivery

Reference Document: SmartTrack PRD v1.1

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Total Duration: 35 Days (Day 1 - Day 35)

Launch Target: Day 35

Accelerated 35-day delivery plan leveraging the existing React web UI prototype (80% complete). All workstreams run in parallel from Day 1. MVP-first approach covers all PRD v1.1 core flows; advanced analytics and Phase 2 roadmap items deferred to post-launch.

1. Executive Summary - 35-Day Sprint Plan

This accelerated plan delivers the complete SmartTrack platform in 35 days (Day 1 - Day 35) by running parallel development workstreams simultaneously, reusing the existing React web UI (already 80% built), and adopting an MVP-first strategy that covers all critical user journeys from PRD v1.1.

Sprint Strategy

- Parallel workstreams: Backend, Web Integration, Mobile (Driver + Parent), Mobile (Admin + Guest)
- Existing web UI saves significant time - focus shifts to API wiring and mobile builds only
- Shared OpenAPI contract published Day 1; all teams integrate against mock server until Day 8
- Daily standups + milestone demos on Day 7, Day 14, Day 21, Day 28
- MVP scope: all core flows live; bulk import, advanced reports, training centre = post-launch v1.1

Current Assets (Head Start)

Asset	Status	Details
Web UI Prototype	DONE	Public site, Super Admin (11 pages), School Admin (16 pages), Auth flows
Mock Data Layer	DONE	Full data models in TypeScript - maps directly to PostgreSQL schema
Component Library	DONE	Tables, charts, maps UI, forms, layout, theme system
Backend API	NOT STARTED	Day 1-14 priority
Flutter Apps	NOT STARTED	Day 1-28 parallel build

2. Day-by-Day Master Schedule

Day 1 - Day 7: Foundation & Kickoff

Squad	Focus	Deliverables	PRD Reference
Backend	DB + Auth	PostgreSQL schema (22 tables), JWT auth, user	PRD Sec.8, Sec.11
Backend	Core APIs	Students, drivers, buses, routes, stops endpoints	PRD Sec.4.2-4.5
Web Squad	API Layer	Axios client, auth interceptors, replace mockData	PRD Sec.2-4
Web Squad	Onboarding	Wire onboarding form + plan selection to live API	PRD Sec.2.1
Mobile A	Driver App Setup	Flutter project, auth, dashboard, FCM token regis	PRD Sec.6.1-6.2
Mobile B	Parent App Setup	Flutter project, auth, home dashboard cards	PRD Sec.7.1-7.2
Mobile C	Admin Mobile	Flutter project, auth, dashboard summary cards	PRD Sec.5.1
DevOps	Infrastructure	AWS staging, CI/CD pipelines, S3 buckets, env conf	

Day 8 - Day 14: Core Features Live

Squad	Focus	Deliverables	PRD Reference
Backend	Trip Engine	Trips, attendance, QR generation (Safety/Route/	PRD Sec.10
Backend	Real-time	Socket.IO setup, bus_locations ingestion endpoint	PRD Sec.4.6
Web Squad	Super Admin	Connect all 11 Super Admin pages to live APIs	PRD Sec.3
Web Squad	School Admin	Students, drivers, buses, routes pages connected	PRD Sec.4.2-4.5
Mobile A	Driver Trips	QR scanner, pickup trip flow (6 steps), GPS track	PRD Sec.6.3
Mobile B	Parent Tracking	Live bus map (5s updates), morning/evening cards	PRD Sec.7.3
Mobile C	Admin Mobile	Live bus map, bus status list, notification centre	PRD Sec.5.2-5.3
QA	Smoke Tests	Auth flows, CRUD operations, API contract validati	

Day 15 - Day 21: Trip Flows & Notifications

Squad	Focus	Deliverables	PRD Reference
Backend	Notifications	FCM push (all 18 triggers), email SMTP, WhatsA	PRD Sec.9
Backend	Transfers	Bus transfer module, guest_trips, guest driver app	PRD Sec.4.14-4.15
Web Squad	Operations	LiveMap Socket.IO, attendance, leave, lost & fou	PRD Sec.4.6-4.11
Web Squad	Messaging	Compose message, notification centre, bus transfer	PRD Sec.4.9-4.14
Mobile A	Driver Drop	Drop trip flow, bus breakdown alert, lost & found	PRD Sec.6.4-6.7
Mobile A	Guest Driver	Guest login, student QR scan, submit for approval	PRD Sec.6B
Mobile B	Parent Features	Leave application, proximity alerts, location pin	PRD Sec.7.4-7.7
Mobile C	Admin Approvals	Leave approval, guest driver approval, bus transfe	PRD Sec.5.5-5.10
QA	Integration	End-to-end pickup trip: driver scan -> parent noti	-

Day 22 - Day 28: Full Integration & Polish

Squad	Focus	Deliverables	PRD Reference
Backend	Remaining APIs	Subscriptions, billing, support tickets, training	PRD Sec.3.3-3.9
Backend	Reports	Attendance export, route performance, revenue rep	PRD Sec.4.13
Web Squad	Final Pages	Subscriptions, reports, guest drivers, support tic	PRD Sec.3-4

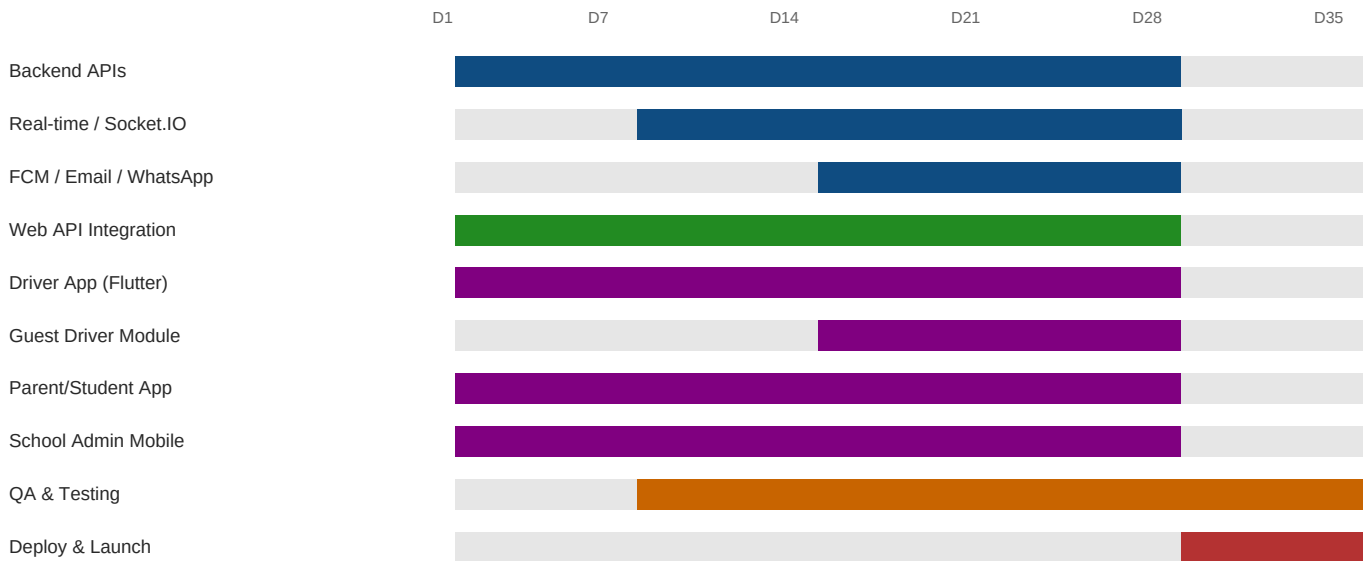
Web Squad	QR + Files	QR download/print, S3 photo uploads, bulk student	PRD Sec.10.3
Mobile A	Driver Polish	Compose message, help/emergency alerts, profile	PRD Sec.6.8-6.10
Mobile B	Parent Polish	Attendance history, lost & found claims, compose message	PRD Sec.7.8-7.11
Mobile C	Admin Polish	Bulk messaging, compose message, support ticket	PRD Sec.5.6-5.9
QA	Full Regression	All user roles, all trip types, notification delivery	-

Day 29 - Day 35: QA, UAT & Launch

Squad	Focus	Deliverables	PRD Reference
All Squads	Bug Fixes	P0/P1 bugs only; feature freeze after Day 30	-
QA	UAT	Pilot school walkthrough: onboard -> trip -> parent	-
QA	Performance	Load test: 50 concurrent buses, 500 parents tracked	-
DevOps	Production	Production deploy, SSL, domain, monitoring dashboard	-
DevOps	App Stores	Play Store + App Store submission (Driver + Parent)	-
PM	Go-Live	Credential delivery to pilot school, training session	-

3. 35-Day Gantt Chart (Parallel Tracks)

All tracks run simultaneously. Bar length = active days.



Blue=Backend | Green=Web | Purple=Mobile | Orange=QA | Red=Launch

4. Development Workstreams

Backend Core	Node.js, PostgreSQL, JWT, all REST APIs, Socket.IO
Integrations	FCM, email SMTP, WhatsApp API, S3, QR generation
Web Integration	Connect existing React UI to APIs, live map, file uploads
Driver + Guest Mobile	Driver Flutter app + guest driver flow
Parent Mobile	Parent/Student Flutter app, live tracking, leave
School Admin Mobile	School Admin mobile app
QA & Testing	Test plans from Day 1, automation, UAT coordination
DevOps	CI/CD, AWS, staging + production environments

5. Deliverables by Layer

5.1 Backend (Node.js + Express + PostgreSQL) - Day 1-28

- Day 1-7: DB schema (22 tables), JWT auth, schools/plans/users CRUD
- Day 1-7: Students, drivers, buses, routes, stops APIs + QR code generation
- Day 8-14: Trip engine (pickup/drop state machine), attendance marking, Socket.IO
- Day 15-21: FCM notifications (18 triggers), email + WhatsApp credential delivery
- Day 15-21: Bus transfer, guest driver approval, lost & found, leave management
- Day 22-28: Subscriptions/billing, support tickets, training modules, reports/export

5.2 Web Frontend (React) - Day 1-28

- Day 1-7: API client setup, auth integration, onboarding flow live
- Day 8-14: Super Admin (11 pages) + School Admin core pages (students, drivers, buses, routes)
- Day 15-21: LiveMap real-time, attendance, leave, messaging, bus transfer, guest drivers
- Day 22-28: Subscriptions, reports, QR print/download, bulk import, file uploads, polish

5.3 Driver Mobile App (Flutter) - Day 1-28

- Day 1-7: Project setup, login, dashboard, FCM registration
- Day 8-14: Safety QR + Route QR scanner, full pickup trip flow with GPS
- Day 15-21: Drop trip flow, bus breakdown alert, lost & found, guest driver module
- Day 22-28: Emergency alerts, compose message, notifications, profile, offline sync

5.4 Parent/Student App (Flutter) - Day 1-28

- Day 1-7: Login, home dashboard (bus status, morning/evening activity cards)
- Day 8-14: Live bus tracking map with 5-second GPS updates and ETA
- Day 15-21: Leave application, proximity alerts, pickup/drop location pin update
- Day 22-28: Attendance history, lost & found claims, student QR, messaging, profile

5.5 School Admin Mobile App (Flutter) - Day 1-28

- Day 1-7: Dashboard, auth, notification badge
- Day 8-14: All buses live map, bus status list with driver info
- Day 15-21: Leave approval, guest driver trip approval, bus transfer from mobile
- Day 22-28: Bulk messaging, compose message, support tickets, final polish

6. Key Milestones

- | | |
|--------------|---|
| Day 1 | Project kickoff; OpenAPI spec started; all repos and environments ready |
| Day 2 | OpenAPI contract frozen; staging database live |
| Day 5 | All apps authenticate; core CRUD APIs functional |
| Day 7 | Phase 1 complete; foundation demo to stakeholders |

Day 10	First end-to-end pickup trip: driver scans QR -> parent gets notification
Day 14	Phase 2 complete; Super Admin + School Admin web connected
Day 15	All 3 mobile apps in beta; real-time GPS tracking live
Day 21	Phase 3 complete; bus transfer + guest driver flows working
Day 28	Phase 4 complete; full platform integrated; feature freeze begins
Day 30	Feature freeze; P0/P1 bug fixes only
Day 33	UAT passed with pilot school
Day 35	Production deployed; app stores submitted; go-live

7. MVP Scope vs Post-Launch (v1.1)

IN SCOPE (Day 1 - Day 35)	POST-LAUNCH (v1.1+)
All auth flows (login, OTP, reset password)	Bulk school Excel/CSV import
School onboarding + Super Admin approval	Advanced analytics & revenue forecasting
Student/driver/bus/route CRUD	Training & Resource Centre module
Pickup + drop trip flows (driver app)	Scheduled message delivery
Live GPS tracking (web + mobile)	Audit logs (full history)
FCM push notifications (core triggers)	Multi-language (Arabic, Malayalam, etc.)
QR codes (Safety, Route, Student)	Facial recognition attendance
Attendance + leave management	In-app threaded chat
Bus transfer + guest driver flows	Driver rating system
Lost & found reporting + claims	White-label branding per school
Parent live tracking + proximity alerts	AI route optimisation
School Admin mobile (core features)	IoT GPS hardware fallback
Compose messaging (admin/driver/parent)	Fleet maintenance module

8. PRD v1.1 Feature Coverage (35-Day Plan)

PRD	Feature	Days	Notes
Sec.2	Public Website & Onboarding	Day 1-7	Web UI done + API Day 1-7
Sec.3	Super Admin Panel	Day 8-28	Web UI done + APIs Day 8-28
Sec.4	School Admin Panel	Day 8-28	Web UI done + APIs Day 8-28
Sec.5	School Admin Mobile	Day 1-28	New Flutter build
Sec.6	Driver Mobile App	Day 1-28	New Flutter build
Sec.6B	Guest Driver Flow	Day 15-21	Within Driver app
Sec.7	Parent/Student App	Day 1-28	New Flutter build
Sec.8	Database Schema	Day 1-3	22 tables on staging
Sec.9	Notifications (18 triggers)	Day 15-21	FCM + email + WhatsApp
Sec.10	QR Code System	Day 1-14	Server-generated HMAC QRs
Sec.11	Common Features	Day 1-28	Auth, offline, profile across all apps
Sec.12	Future Roadmap	Post-launch	Phase 2-4 out of scope

9. Critical Success Factors

- Existing web UI prototype eliminates significant frontend build time
- OpenAPI contract frozen by end of Day 2 - no API changes after Day 8
- Feature freeze at Day 30 - only P0/P1 bug fixes on Day 31-35
- Pilot school identified before Day 1 for UAT on Day 33
- Google Maps SDK keys, FCM project, WhatsApp Business API pre-approved before Day 1
- Daily standups + integration demo on Day 7, Day 14, Day 21, Day 28

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Sprint Duration: Day 1 - Day 35